What Survives Compression and What Does Not: An In-Silico Audit of Connectome-Based Mind-Migration Proposals Using a Functional Preservation Score

[Author(s) withheld for review]

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Abstract

Speculative proposals for "mind migration" describe reading a brain's identity, compressing it, and installing it into a new substrate. Such proposals assemble several active research areas (connectomics, graph neural networks, memory prosthetics, neural interfaces, optogenetics) into a single pipeline, typically relying on the most optimistic reading of each. We examine the computational core of this pipeline directly, using an auditable procedure on public data that runs on a single laptop. The study is organised around a Functional Preservation Score (FPS): a null-normalised, behaviourally grounded measure of how much of a neural system's input–output function survives an operation such as compression, alignment, transport, or writing. We apply FPS across five pre-registered studies on the *C. elegans* connectome, whole-brain calcium activity [8], and hippocampal replay (DANDI 000044).

Two of the studies yield positive characterisations and three form a negative result. First, low-bit quantisation of a real connectome runs into a structure-dependent sparsity floor: uniform int4/ternary quantisation is forced to roughly 73%/98% effective sparsity because the heavy-tailed integer synaptic weights round to zero, whereas mixed-precision protection of hub neurons lowers these floors to about 27%/32%. Second, decoded hippocampal replay sequences compress heavily while remaining identifiable: a 48-unit × 25-bin sequence (∼38 kbit) stays identifiable at a cross-validated top-1 accuracy of 0.69 (95% CI $[0.63, 0.74]$; chance 0.125; permutation $p < 10^{-3}$) from latents on the order of 10–50 bits per event, and a bounded "background-sync" store shows a graceful-versus-cliff trade-off between capacity and interference. Third, the migration pipeline fails in simulation at each stage, for largely independent reasons. The putative "identity payload" is not a functionally separable module (permutation $p \geq 0.16$ at every tested size). Cross-individual functional transport collapses once the target's actual wiring is respected (fidelity $1.00 \to 0.40$, 95% CI $[0.31, 0.49]$, $p < 10^{-3}$) or the alignment is imperfect (0.06, $[0.00, 0.13]$). And the proposed biologically plausible write mechanism cannot reach the function the target topology provably admits (plausibility gap 0.45, 95% CI $[0.27, 0.64]$, $p < 10^{-3}$, $d = 2.0$).

Taken together, compression of neural data is a quantifiable phenomenon, while migration of function across substrates is not supported by the proposed pipeline on the systems we can currently test. We release FPS, the dataset loaders, and the full set of operations as a measurement toolkit.

Scope. All results are obtained on a 306-neuron nematode connectome, small recurrent networks, and reflex- or trajectory-scale functions. They say nothing about reading or writing a living human brain. The biological prerequisites for the full pipeline remain absent, and we make no claim about them; the studies test coherence in simulation, not biological feasibility.

1 Introduction

A recurring theme in recent discussions of biological aging is to treat it as a reversible information-processing problem. Mind migration is one speculative endpoint of that view. In the version we examine, a brain is partitioned into a regenerable "Base-OS" (autonomic, sensory, and motor circuitry) and a transferable "identity payload" (association cortex and hippocampus); the payload is then compressed through sparsification and low-bit quantisation, aligned to a differently wired target brain, and installed through activity-dependent plasticity. Each of these steps draws on a real experimental result, but the composition has never been tested as a whole, and several of the links have no supporting evidence at any scale.

We do not attempt the biological pipeline, which is infeasible with current or foreseeable tools. Instead we treat preservation of function as a measurable quantity and ask which stages of the computational pipeline can actually be built, and where and why the remaining ones fail. Posing the question this way turns an otherwise unfalsifiable goal into a measurement problem that admits decision-relevant evidence from public data on ordinary hardware.

The paper makes five contributions. (1) We define FPS, a substrate-agnostic measure of functional preservation with an explicit trivial-baseline null (§3.1). (2) We characterise compression of a real connectome and identify a quantisation sparsity floor (§4). (3) We characterise the compressibility of real hippocampal replay sequences and describe a capacity–interference relationship for bounded storage (§5). (4) We provide a three-stage in-silico test of the partition–align–write pipeline and locate the failure mode at each stage (§6). (5) We release a reproducible, laptop-scale toolkit (§8).

2 Related work and claim provenance

We checked the frontier results that the pipeline depends on against their primary sources. Most are genuine, but several are mis-attributed or overstated in popular accounts. On synaptic turnover and memory, torpor is known to shed roughly 50–65% of synapses with memory preserved [12], which supports sparsification that is structured rather than uniform. On connectome-driven models, FlyWire [3] and flyGNN [4] show that a connectome can drive simulated behaviour, but only once the dynamical parameters are fitted; the topology alone does not suffice. On memory prosthetics, the multi-input multi-output devices of Hampson et al. [6] improved human recall by about 35–37%, but this is restoration within a single subject rather than transfer of identity between subjects. On neural interfaces, high-density wireless arrays [7] record from the subdural surface and do not read at synaptic resolution. On alignment, Gromov–Wasserstein optimal transport aligns brains at the scale of functional parcels [11, 15], but it is NP-hard and has not been tested for functional transport on connectomes.

Generic graph-network pruning and quantisation [9] and scalable Gromov–Wasserstein [14] are established methods. What is new here is their composition on connectome substrates under a functional, behaviourally grounded metric, together with the explicit test of the migration stages that have no empirical basis.

3 Methods

3.1 Functional Preservation Score

Let a system map a fixed battery of stimuli to outputs. For a reference system with outputs Y_{ref} and a transformed system with outputs Y, define an R^2-style fit $\text{raw}(Y) = 1 - \|Y - Y_{\text{ref}}\|^2 / \|Y_{\text{ref}} - \overline{Y_{\text{ref}}}\|^2$.

The score is normalised by the best trivial baseline, the larger of a shuffled stimulus–response mapping and a predict-the-mean baseline:

$$\textsc{fps} = \mathrm{clip}\left(\frac{\mathrm{raw}(Y) - \mathrm{raw}(Y_{\mathrm{null}})}{1 - \mathrm{raw}(Y_{\mathrm{null}})},\, 0,\, 1\right). \tag{1}$$

An FPS of 1 means the transformed system reproduces the reference function; an FPS of 0 means it performs no better than the best trivial control. Normalising by the null matters in practice, because without it a system that always produces a plausible-looking but input-independent output can score deceptively well. For identity-style tasks we also use identification accuracy, which asks whether reconstruction i is nearest to its own original among all events; it is bounded and has an exact chance level $1/n$.

3.2 Substrates and data

We use the White/Cook *C. elegans* somatic connectome [1, 13] via the OpenWorm ConnectomeToolbox (306 neurons, 3,158 edges; 81 sensory, 95 interneuron, 130 motor). Dynamics are a leaky-rate tanh recurrent model reading out motor neurons over a fixed low-pass stimulus battery. Whole-brain calcium activity is from Kato et al. [8] (5 worms, ∼109–135 neurons). Hippocampal replay is from DANDI 000044 [2, 5] (Buddy session, 48 pyramidal units plus linearised position, streamed without full download), with a nelpy CA1 linear-track session for the structure gate.

3.3 Operations and statistics

Compression uses magnitude pruning (uniform) versus engram-aware pruning that protects high-degree, betweenness, or k-core hubs, with symmetric quantisation to int8/int4/ternary and an optional mixed precision that keeps hubs at higher bit depth; policies are compared at matched total effective sparsity. Alignment uses a self-contained entropic Gromov–Wasserstein solver (square loss with Sinkhorn projection [10]); transport pushes weights through a node correspondence under a Peters'-rule feasibility mask. The write stage uses a fixed-topology rate host with strictly local three-factor plasticity (audited: no backprop, no per-weight target gradient) and a global least-squares optimiser as an upper bound.

All headline numbers are reported as a point estimate with a percentile bootstrap 95% confidence interval (10,000 resamples) over the relevant randomisation, with the sample size stated. Comparisons carry a one-sided permutation test (10,000 permutations, add-one-corrected) and, where applicable, Cohen's d. To avoid selection bias we report a pre-registered primary engram policy (strength) rather than the maximum over variants. The identification frontier uses 30-split repeated cross-validation with the autoencoder fit on the training split only.

4 Study 1: Connectome compression

We ask how far a connectome-driven model can be compressed, along sparsity, bit depth, and structure, before its function collapses, and whether structured sparsification outperforms uniform pruning. At matched effective sparsity under int4, using the pre-registered primary policy and eight stimulus seeds, the engram-aware policy preserves function better than uniform pruning: FPS of 0.30 (95% CI $[0.12, 0.48]$) versus 0.03 ($[0.00, 0.09]$) at target sparsity 0.5 ($p = 0.015$, $d = 1.3$), with comparable margins at 0.6 and 0.7.

The central finding is the floor (Fig. 1). Uniform quantisation cannot reach low sparsity on the real connectome, because the heavy-tailed integer weights round to zero: the effective-sparsity floor

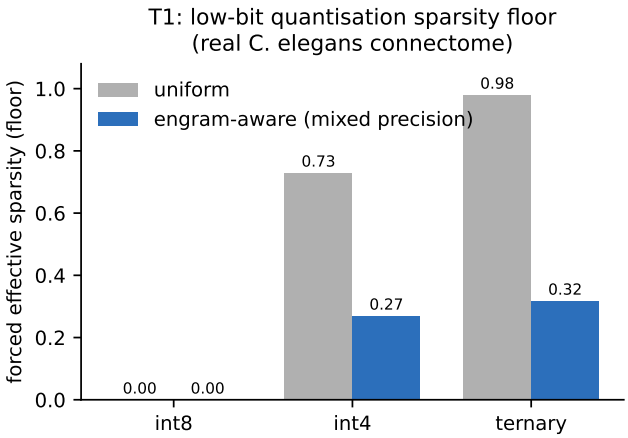


Figure 1: Low-bit quantisation imposes an effective-sparsity floor on the real *C. elegans* connectome under uniform quantisation (grey); mixed-precision hub protection (blue) lowers it. The floors are deterministic weight-distribution properties.

is 0.73 for int4 and 0.98 for ternary under uniform quantisation, against 0.27 and 0.32 under engram-aware mixed precision. At int8 there is no floor. These floors are deterministic properties of the weight distribution. The structured advantage, by contrast, is not robust: in a clean int8 comparison it depends on regime and does not reproduce under an independent sigmoid-conductance model, so we drop the broader claim and retain only the floor.

5 Study 2: Replay-sequence compression

We ask whether decoded hippocampal replay sequences can be compressed to few bits while preserving their content. Place-ordered run-epoch sequences first show decodable trajectory structure (Bayesian decode score 0.34 versus a time-shuffled null of 0.20 across 62 sequences). Under 30-split cross-validation with the autoencoder fit on the training split only, the uncompressed model identifies held-out replay events at top-1 0.69 (95% CI $[0.63, 0.74]$) against chance 0.125 (permutation $p = 10^{-4}$). Identification degrades gradually with the bit budget, with non-overlapping intervals, so a 48×25 sequence ($\sim$38 kbit raw) remains well above chance down to roughly 8–16 bits per event, a compression ratio of order 10^3 (Fig. 2, left). Streaming events into a fixed-size store (Fig. 2, right) shows a genuine capacity limit: a superposition store degrades gradually while a slot store is perfect to capacity and then drops sharply. These results make the compressed-backup idea plausible as compression of decoded sequences; they do not show that such a backup contains a memory.

6 Study 3: The migration pipeline in simulation

Each stage is a surrogate test of a biological stage for which no direct evidence exists; the question is whether the stage is coherent in simulation on the nearest testable substrate.

Partition separability. Holding the sensory and motor interface fixed, we test whether ablating the asserted interneuron payload degrades the input–output transformation more than ablating a random interneuron subset of the same size. It does not. Against a 500-sample null, the asserted payload is not significantly more disruptive at any tested size (permutation $p = 0.17, 0.16, 0.27, 0.37$

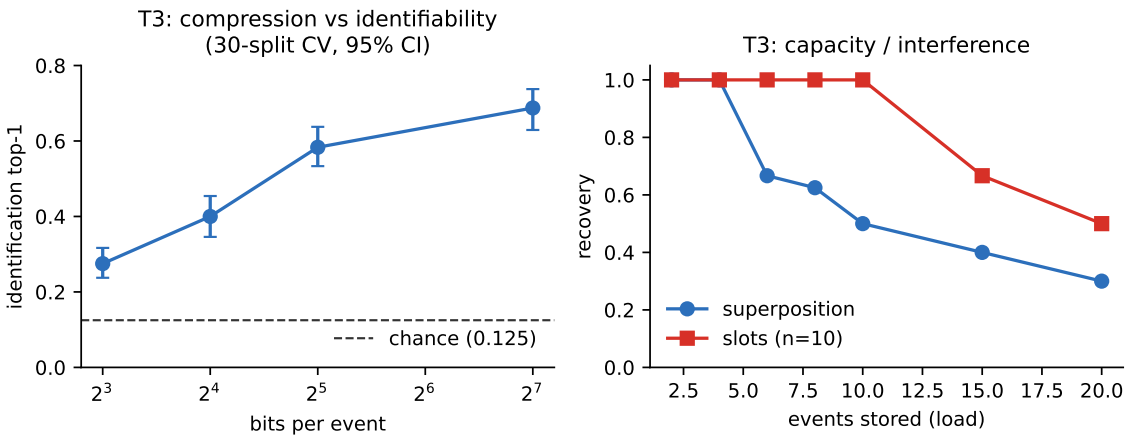


Figure 2: Left: identification accuracy versus latent bit budget under 30-split cross-validation (95% CI); dashed line is chance. Right: recovery from a fixed-size incremental store versus load, for a superposition store and a slot store.

at payload fractions 0.1–0.4). We find no significantly separable identity module; the function is distributed.

Functional transport. With a second individual generated as a controlled-divergence perturbation of the first, entropic Gromov–Wasserstein recovers 0.76 (95% CI $[0.74, 0.78]$) of the ground-truth correspondence at 5% rewiring. The transport, however, does not survive realistic conditions (Fig. 3). Given an oracle correspondence and unconstrained writing, fidelity is 1.00; once the transported weights must respect the target's wiring, it falls to 0.40 (95% CI $[0.31, 0.49]$, $p = 3 \times 10^{-4}$); using the estimated correspondence it is 0.06 ($[0.00, 0.13]$, $p = 2 \times 10^{-4}$). High correspondence accuracy does not imply preserved function; the binding constraint is the target's own wiring.

Local-rule writing. Across ten seeds, a global optimiser reaches a ceiling of 1.00, confirming the topology can host the function, while strictly local reward-modulated plasticity reaches only 0.55 (95% CI $[0.36, 0.73]$), leaving a plausibility gap of 0.45 ($[0.27, 0.64]$, $p = 10^{-4}$, $d = 2.0$; Fig. 4). Ripple-timed gating confers no advantage (the ripple-minus-continuous difference has a CI including zero).

The three stages together. The stages fail for largely independent, compounding reasons: there is no clean module to extract; even the whole graph cannot be placed once alignment is imperfect and the target's wiring is respected; and even where the topology can host the function, the proposed write mechanism cannot reach it. The target substrate's own wiring is the limit common to the last two stages.

7 Discussion

The computational core of the pipeline separates into two parts. Compression of neural data is measurable and bounded (Studies 1 and 2). Migration of function across substrates is not supported by the proposed partition–align–write pipeline (Study 3), with the target substrate's wiring as the recurring obstacle. Several limitations qualify these conclusions. The work is at small scale (a

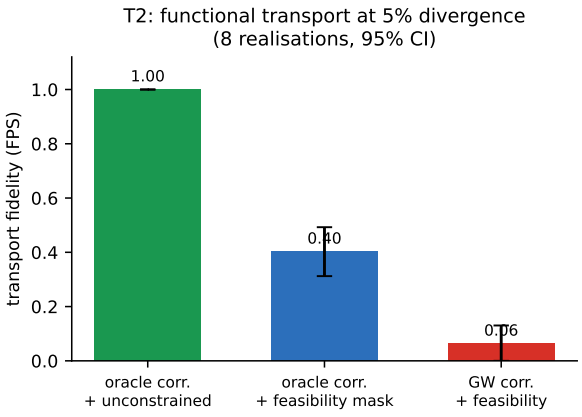


Figure 3: Transport fidelity at 5% divergence. The feasibility mask and imperfect alignment each collapse fidelity relative to the oracle/unconstrained ideal.

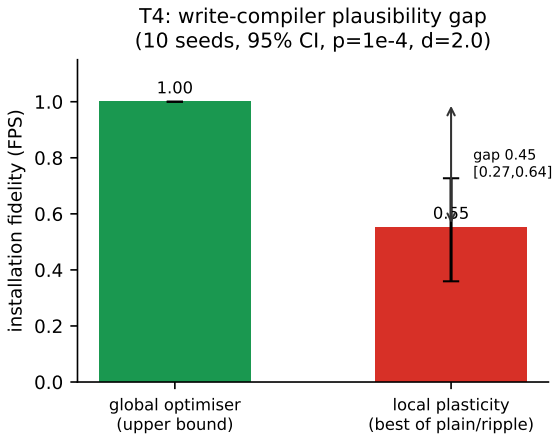


Figure 4: The write-compiler plausibility gap: a global optimiser succeeds where local plasticity does not, on the same feasible topology.

306-neuron connectome, activity recordings of at most 135 neurons, and 40-neuron networks); the dynamics are simple rate models; some test sets are small; the second individual in Study 2 is a divergence model of a single real connectome; and FPS is a functional measure, not a claim about subjective experience. The biological prerequisites for the full pipeline remain absent and are not addressed. All work is computational on public animal datasets, with no human subjects, wet-lab, organoid, or cloning work.

8 Reproducibility

All results are deterministic and run on a laptop. Each claim maps to a committed JSON artifact and the script that produced it. Data are fetched via documented scripts; the metric, loaders, and operations are released, and a test suite covers the substrate contracts. Statistically hardened artifacts (bootstrap CIs and permutation p-values) accompany each headline claim.

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